



Name:

Deep Learning Computer Assignment 1

STN:

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1. In this question, we want to train and evaluate a single-layer perceptron network on a training dataset. Open the file `simplifiedataClassification.m`. In this program, the data are first prepared. The data consist of 100 two-dimensional samples generated in three classes. In this exercise, the first 70 data points are considered training data. The network weights are initialized with values drawn from a Gaussian distribution. First, try to update these weights using the perceptron learning rule. Then, considering the rest of the code, explain how the network operates.
 2. In this question, the data are generated in a somewhat more complex form. Investigate whether these data can be classified using a two-layer perceptron network. Finally, implement an appropriate perceptron network, train it, and discuss the network's performance on these data. For this simulation, you may use the partially prepared code `m.moondataClassification`.

Thank you

Good luck

Note 1: Upload the program codes to the system as separate files.

Note 2: Prepare a report of the code execution and the requested questions, and upload it to the system as a PDF file.

Note 3: This is an individual assignment and copying is not permitted. However, you may consult with one another regarding coding experience and the analysis and comparison of results.

Note 4: You may use any programming language, but the use of ready-made toolboxes is prohibited.